Predicting Executive Functioning in At-Risk Children: Both Mothers and Father-Figures Matter

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Introduction

- **Executive Function (EF)**: higher level cognitive skills (e.g., mental flexibility, inhibitory control, working memory) that allow for goal-directed behaviors.
  - High EF in preschoolers related to later school achievement and social skills (Mischel et al., 2011).
  - Low SES Children at risk for EF deficits (Hackman et al., 2015).
- **Caregiver interactions** are crucial for learning self-regulation:
  - *Attachment theory*: children learn to self-regulate within secure relationships via interactions present in *secure base-safe haven cycles* (Bowlby, 1988).
  - Research (e.g., Carlson, 2003) indicates various aspects of mother parenting are related to EF.
  - Little research has looked at the impact of father figures on EF, although father involvement has been positively related to other cognitive outcomes (e.g. Cabrera et al., 2000).
  - Researchers have not examined the pathways from mother parenting, father support, and child EF in an at risk sample

Research Question & Hypotheses

How is mother parenting and support from father-figures related to EF development in at-risk children?

Hypotheses

- Quality of maternal parenting in early childhood and level of support from father-figures in both early and middle childhood will have direct effects on child EF
- Paths from mother parenting and father support to child EF will vary depending on attachment status

Methods

Participants

- Minnesota Longitudinal Study of Risk and Adaptation (Egeland & Brunnquell, 1979).
- N = 182, 97 males and 85 females
- High risk at birth: 100% at or below poverty level, young mothers (M = 20.7), 60% single mothers

Measures

Child EF: Early Childhood Lab Composite

- Barrier Box (42 mo)
- Curiosity Box, Gift Delay, & Ego Control Rating (54 mo)

Child EF: Middle Childhood School Composite

- EF-related questions from Teacher completed Deveraux K-3rd grades

Mother Parenting:

- 42 months video-taped teaching task; coded for:
  - supportive presence,
  - quality of instruction,
  - respect for autonomy,
  - structure and limit setting
- Attachment classification measured at 12 and 18 mo
  - always secure versus insecure at one or more time points

Support from Father-Figures:

- Coded for level of emotional support to child from 18 interviews across childhood
- Rated on a 4 point scale:
  - 1=No male in home
  - 2 = Low quality support
  - 3 = Moderate/Average support
  - 4 = High quality support
- Scores given for Early Childhood (EC) and Middle Childhood (MC). Any and all males in the home during time period included in rating.

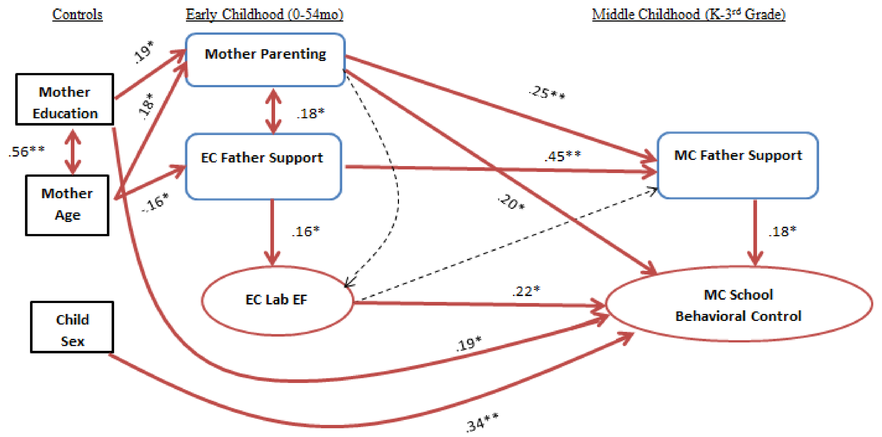
Results

Bivariate Correlations

	1	2	3	4	5	6	7	8
1. Lab EF	--	.253	**.441****	**.402****	**.367****	.098	**.273***	-.078
2. School Behavioral Control	.179	--	**.319***	.202	-.123	.129	.224	.224
3. Mother Parenting	-.024	.176	--	**.287***	-.136	**.429****	**.476****	-.023
4. EC Father Support	.131	-.038	-.001	--	.118	.015	.217	-.121
5. MC Father Support	.028	**.256***	**.392****	**.398****	--	-.035	-.002	.100
6. Mother Age	.194	-.084	.195	-.040	-.029	--	**.508****	.057
7. Mother Education	.192	.117	.108	-.033	.116	**.604****	--	-.056
8. Child Sex	.214	**.342****	-.115	-.032	.042	-.059	-.067	--

*p < .05. **p < .01. Bottom left (shaded): children who were insecure at one or more time points. Top right: children who were always secure.

Best Fitting Path Model



Model Fit Stats	χ^2	df	χ^2/df	$\Delta\chi^2$(df)	CFI	RMSEA
3 (vs. 2) Final Model	.608	2	.304	7.903(2)*	1.00	.000

This model accounted for 7.4% of EC Lab EF Variance and 30.9% of MC School Behavior Control Variance.

Moderation Analysis

- The pathway from Mother Parenting to EC Lab EF was moderated by attachment, being significant only for secure children.
- The moderated model was a better fit than the fully constrained model ($\Delta\chi^2$(df) = 5.30(1), p < .05) and for secure children, 23% of variance in EC Lab EF was then accounted for.

Discussion

- Hypotheses were generally supported:
  - Both **quality of mother parenting** and **level of male support** were related to executive function skills and school behavioral control.
  - **Mother parenting** was a significant predictor of early childhood EF for securely attached children only.
- Strengths of the study:
  - Included all father-figures in the child's life during relevant time periods
  - Data from multiple sources, including rich observational measures, laboratory tasks, and reports from mothers and teachers.
- Overall, we found that both mother parenting and support from father figures are important for the development of executive function in at-risk children across the preschool and grade school years.

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